

Resume Tailoring Platform HLD

Phase 1 Prototype | Version 1.0 | April 2026

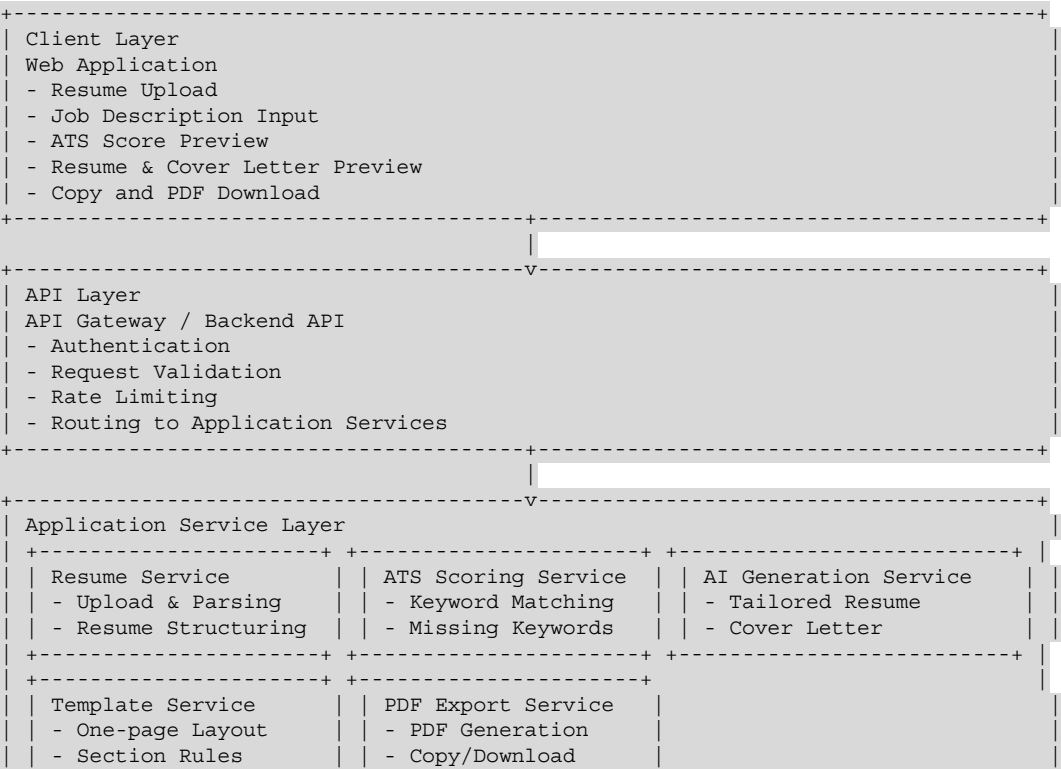
Change Log

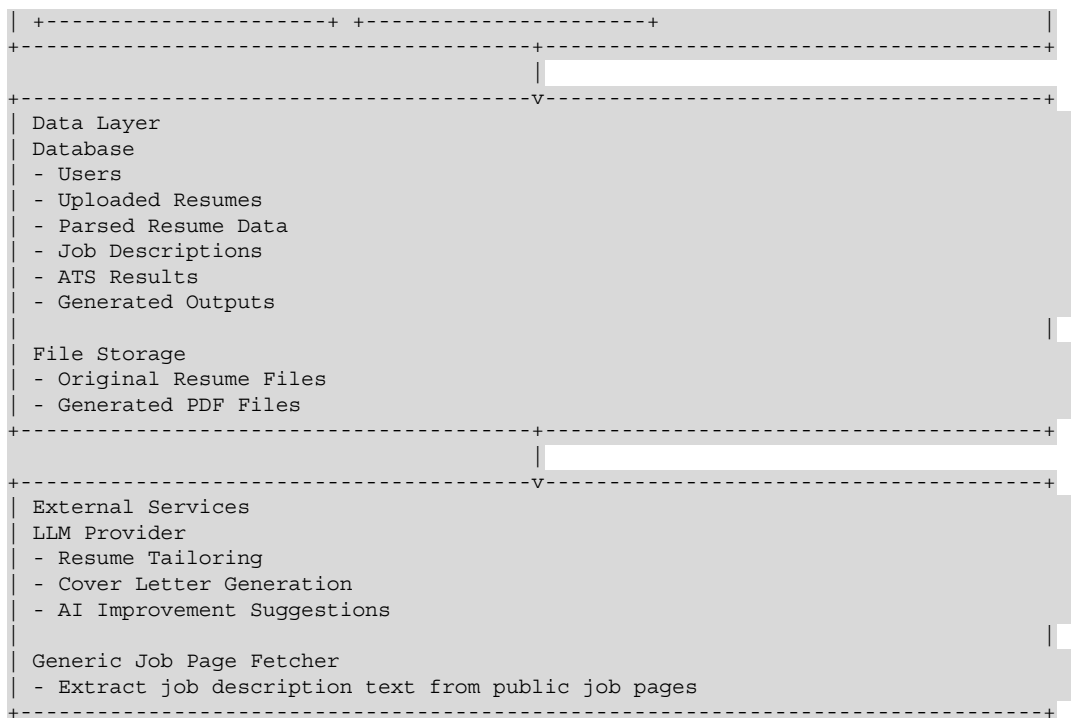
Version	Date	Changes
1.0	2026-04-30	Initial HLD draft
1.1	2026-04-30	Completed full HLD structure

1. Architecture

1.1 High-Level Architecture Diagram

This section describes the high-level architecture of the Resume Tailoring Platform. The system allows users to upload an existing resume, provide a job description, receive an ATS score, generate a tailored one-page resume, generate a cover letter, and export the final output as PDF.





1.2 Component Overview

This section provides an overview of the main system components and their responsibilities within the Resume Tailoring Platform.

Component	Responsibility	Notes
Web Application	Provides user interface for resume upload, ATS preview, and resume generation.	Main user-facing layer.
Resume Service	Handles resume upload, parsing, and structured extraction.	Converts CV into structured format.
Job Description Service	Accepts JD input or generic job page content.	No site-specific scraping.
ATS Scoring Service	Calculates ATS score and missing keywords.	Fast response required.
AI Generation Service	Generates tailored resume and cover letter.	AI should not invent experience.
Template Service	Ensures one-page resume layout consistency.	Fixed structure enforcement.
PDF Export Service	Generates downloadable PDF output.	Prevents layout overflow.

1.3 Key Architectural Decisions

This section explains the main architectural decisions for the Resume Tailoring Platform and why they were selected for the Phase 1 Prototype.

Decision	Rationale
Web Application First	The product is focused on resume upload, JD input, ATS scoring, resume generation, and PDF export. A web application is simpler and more suitable for MVP than a browser extension.
Keyword-Based ATS Scoring	Provides fast and consistent ATS score calculation within the required 1–2 seconds.
AI as Generation Support	AI is used to improve wording, generate tailored content, and provide suggestions, but it must not invent user experience or fake achievements.
Fixed Resume Template	Ensures consistent structure: Summary, Skills, Experience, Projects, Education.
One-Page Layout Enforcement	The system must control content length and formatting to prevent resume overflow.
PDF Export from Controlled Template	PDF generation should use a fixed layout to ensure professional output and prevent formatting issues.
Generic Job Page Support	MVP avoids site-specific scraping and focuses on pasted JD text or generic public job page extraction.

1.4 Data Flow Principles

This section explains how data moves through the Resume Tailoring Platform, from resume upload to ATS scoring, AI generation, and final PDF export.

Principle	Description
Single Resume Upload	Users upload their resume once. The system stores both the original file and parsed structured data.
Structured Resume Parsing	Uploaded resumes are converted into structured sections such as Summary, Skills, Experience, Projects, and Education.
Job Description Input	Users provide a job description through pasted text or a generic public job page URL.
ATS Scoring Pipeline	The system compares resume data with the job description to calculate ATS score and identify missing keywords.
AI Generation Flow	AI uses structured resume data and ATS insights to generate a tailored resume and cover letter.
Controlled Output Generation	Generated content is passed through a fixed template to guarantee one-page layout consistency.

1.5 Technology Stack

This section defines the recommended technology stack used to implement the Resume Tailoring Platform MVP.

Layer	Technology	Purpose
Frontend	Next.js / React	User interface and resume interaction flow.
Backend API	Python + FastAPI	Core business logic and ATS processing.
Database	PostgreSQL	Stores users, resumes, ATS results, and exports.
File Storage	AWS S3	Stores uploaded resume files and generated PDFs.
AI Layer	OpenAI API	Resume tailoring and cover letter generation.
PDF Engine	HTML to PDF Renderer	Creates downloadable one-page PDF resumes.
Authentication	JWT Authentication	Protects user sessions and API access.

2. User Journey and Flows

2.1 Complete Resume Tailoring Journey

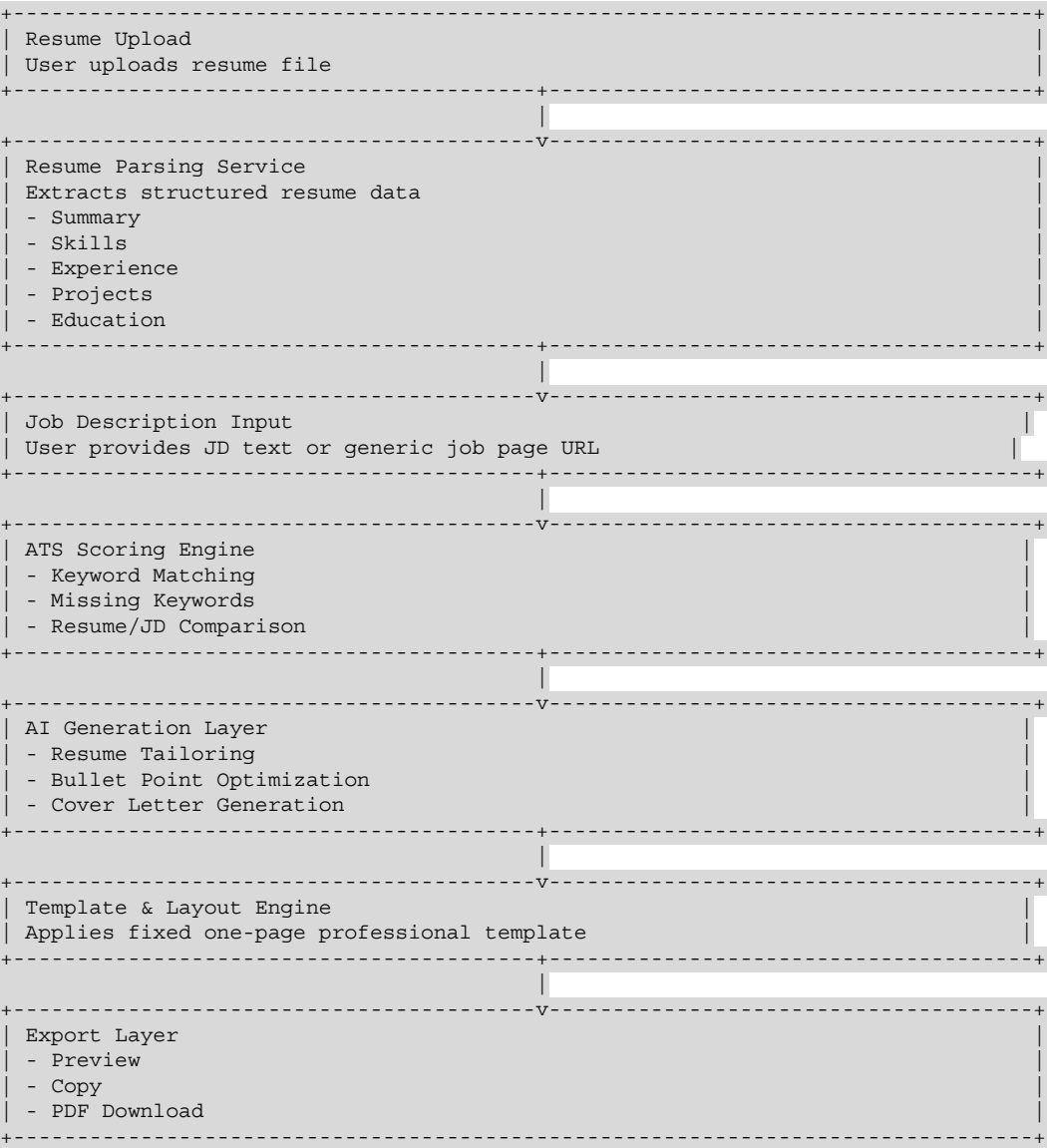
This section describes the end-to-end user journey from resume upload to ATS scoring, tailored resume generation, cover letter generation, and PDF export.

Step	User Action	System Output
1	User opens the web application.	Platform homepage and resume upload interface are displayed.
2	User uploads an existing resume.	Resume file is stored and sent to Resume Parsing Service.
3	System parses the uploaded resume.	Resume data is converted into structured sections.
4	User provides a Job Description (JD).	JD text is extracted and normalized for ATS comparison.
5	ATS analysis starts.	System calculates ATS score and identifies missing keywords.
6	User reviews ATS feedback.	Missing skills, keywords, and recommendations are displayed.

7	User generates a tailored resume.	AI creates an optimized one-page resume.
8	User generates a cover letter.	AI creates a cover letter aligned with the resume and JD.

2.2 Resume Tailoring Workflow

This workflow explains how resume data moves through ATS scoring, AI generation, template formatting, and final export.



3. Service Layer Design

This section describes the core backend services responsible for resume processing, ATS scoring, AI generation, template formatting, and export functionality.

3.1 Resume Service

The Resume Service is responsible for receiving uploaded resumes, storing original files, parsing resume content, and converting unstructured resume data into structured sections.

Capability	Description
Resume Upload	Accepts PDF or DOCX resume files uploaded by the user.
Resume Storage	Stores original uploaded resume files for reuse.
Resume Parsing	Extracts resume content into structured sections.
Structured Data Output	Produces normalized data used by ATS and AI services.
Resume Validation	Checks file type, size, and content quality.

3.2 ATS Scoring Service

The ATS Scoring Service evaluates resume relevance against the provided Job Description by calculating keyword match score, identifying missing skills, and generating structured ATS insights.

Capability	Description
Keyword Matching	Compares resume content against JD keywords.
ATS Score Calculation	Produces a relevance score between resume and job description.
Missing Keyword Detection	Identifies important skills or terms missing from the resume.
Section Analysis	Analyzes strengths and weaknesses across resume sections.
Fast Response Pipeline	Returns ATS score within 1–2 seconds.
ATS Insights Output	Provides structured recommendations for resume improvement.

3.3 AI Generation Service

The AI Generation Service uses structured resume data, Job Description context, and ATS insights to generate a tailored resume, optimized bullet points, improvement suggestions, and a cover letter.

Capability	Description
Resume Tailoring	Rewrites and adjusts resume content to better match the provided JD.
Bullet Point Optimization	Improves bullet points to be clear, impact-based, and metric-oriented where possible.
AI Suggestions	Suggests missing skills, stronger wording, and possible resume improvements.
Cover Letter Generation	Generates a cover letter aligned with the resume and job description.
Content Safety Rules	AI must not invent fake experience, companies, education, or achievements.
Structured Output	Returns generated content in fixed sections required by the resume template.

3.4 Template & Layout Service

The Template and Layout Service ensures that generated resumes follow a fixed professional structure and always remain within a one-page layout constraint.

Capability	Description
Fixed Resume Structure	Applies consistent resume sections such as Summary, Skills, Experience, Projects, and Education.
One-Page Enforcement	Limits content length to prevent resume overflow beyond one page.
Section Prioritization	Determines which sections receive more visual space based on relevance.
Layout Normalization	Ensures spacing, margins, and formatting remain consistent.
Overflow Prevention	Automatically shortens or compresses content when page limits are exceeded.
Professional Styling	Uses ATS-friendly typography and layout rules.

3.5 PDF Export Service

The PDF Export Service converts generated resume content into downloadable PDF format while preserving layout consistency and one-page structure.

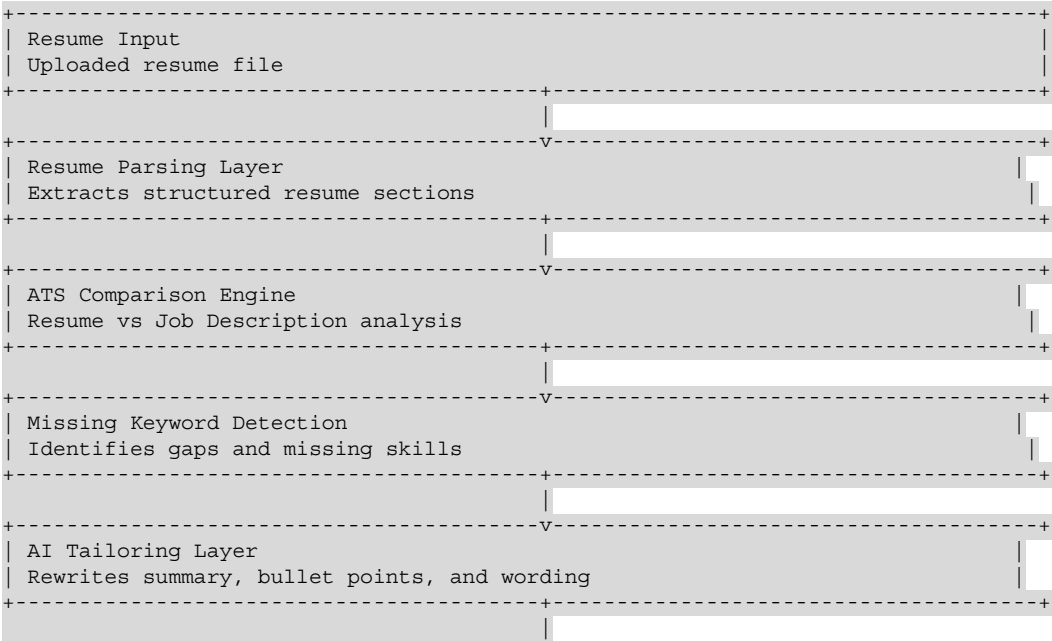
Capability	Description
PDF Generation	Converts final resume layout into downloadable PDF format.
Layout Preservation	Ensures PDF output matches the previewed resume layout.
One-Page Validation	Verifies that exported PDF remains within a single page.
Copy Export	Allows users to copy generated resume content directly.
File Naming	Applies consistent naming for generated resume exports.
Download Management	Handles secure PDF download delivery to the user.

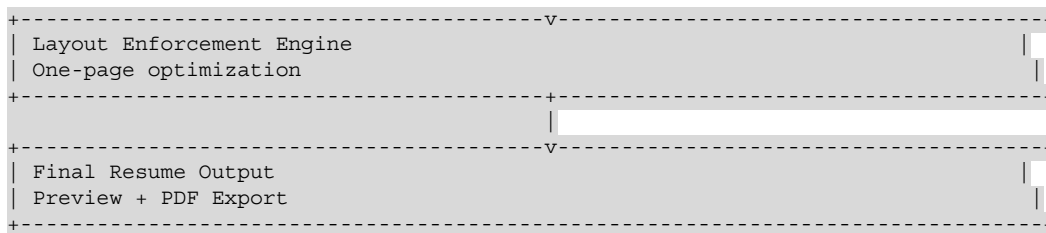
4. Resume Tailoring Engine

This section explains how the Resume Tailoring Engine analyzes resume content, ATS results, and job description context to generate a professional one-page tailored resume.

4.1 Tailoring Pipeline

The tailoring pipeline defines the sequence of processing steps used to transform an uploaded resume into an optimized one-page output.





4.2 ATS Matching Logic

This section explains how ATS scoring evaluates resume relevance against a Job Description using structured keyword matching and weighted scoring.

Logic Rule	Description
Keyword Matching	Resume keywords are compared against job description keywords.
Weighted Skill Scoring	Important JD keywords receive higher scoring priority.
Section-Based Analysis	Resume sections are analyzed independently for relevance.
Missing Keyword Detection	Critical missing skills are highlighted for improvement.
Score Normalization	ATS score is normalized to a consistent percentage scale.
Fast Evaluation Pipeline	Scoring should complete within 1–2 seconds.
AI Enhancement Layer	AI suggestions are applied after ATS scoring is completed.

4.3 One-Page Optimization Logic

This section explains how the system keeps every generated resume strictly within a one-page layout while preserving the most relevant information for the target job.

Optimization Rule	Description
Fixed Section Order	Resume always follows the same structure: Summary, Skills, Experience, Projects, Education.
Content Prioritization	Most relevant JD-matching content receives higher priority in the final resume.
Bullet Compression	Long bullet points are shortened while preserving impact and meaning.
Low-Relevance Removal	Less relevant details may be removed if the resume exceeds one page.
Layout Validation	Generated resume is checked against page height and formatting constraints.

Retry Generation	If the resume overflows, the system regenerates a shorter version.
PDF Preview Check	Final preview must match the exported one-page PDF.

5. Integration Points

This section describes how the Resume Tailoring Platform integrates with external services required for AI generation, storage, parsing, and export.

5.1 External Service Integrations

Service	Purpose	Integration Type
OpenAI API	Generates tailored resume content, cover letters, and AI suggestions.	REST API
AWS S3	Stores uploaded resumes and generated PDF files.	Cloud Storage
Resume Parsing Engine	Extracts structured content from uploaded resume files.	Internal Service
PDF Generation Engine	Converts structured resume output into downloadable PDF.	Internal Library
Generic Job Page Extractor	Extracts job description text from public URLs.	HTTP Request
Authentication Provider	Handles user login and access control.	JWT Authentication

6. Data Model

This section describes the core database entities used to store users, resumes, ATS results, generated outputs, and job descriptions.

6.1 Core Entities

Entity	Purpose	Key Fields
User	Stores user profile and account information.	user_id, email, created_at
Resume	Stores uploaded resume metadata and parsed content.	resume_id, user_id, file_url, parsed_data

Job Description	Stores uploaded or extracted JD text.	jd_id, user_id, source_url, jd_text
ATS Result	Stores ATS score, keyword matches, and insights.	ats_id, resume_id, score, missing_keywords
Generated Resume	Stores AI-generated tailored resume output.	generated_resume_id, resume_id, final_content
Cover Letter	Stores generated cover letter content.	cover_letter_id, resume_id, content
Export Record	Stores exported PDF history and download metadata.	export_id, user_id, pdf_url, created_at

7. Performance & SLA

This section defines performance targets and response time expectations for the Resume Tailoring Platform.

7.1 Performance Targets

Operation	Target Response Time	Notes
Resume Upload	1–2 seconds	Upload confirmation should appear immediately.
Resume Parsing	1–2 seconds	Structured extraction should complete quickly.
ATS Score Calculation	1–2 seconds	Matches customer requirement for fast ATS feedback.
AI Resume Generation	2–4 seconds	Includes tailored resume generation.
Cover Letter Generation	2–4 seconds	Should complete alongside resume generation.
PDF Export	1–2 seconds	Export must preserve formatting.
Full Pipeline Completion	3–4 seconds	Total end-to-end workflow time.

8. Security & Constraints

This section defines system security rules, user protection, and operational constraints for the Resume Tailoring Platform.

8.1 Security Principles

Security Rule	Description
User Authentication	Users must authenticate before uploading resumes or generating content.
Secure File Upload	Uploaded resume files must be validated for type and size.
Private Resume Storage	User resumes and generated outputs remain private to the account owner.
JWT Authentication	Secure API access is protected through JWT-based authentication.
AI Content Restrictions	AI cannot generate fake work experience or false claims.
Rate Limiting	API requests should be limited to prevent abuse.
Temporary Processing	Resume processing should avoid storing unnecessary temporary data.

9. MVP Scope

This section defines the functionality included in the first MVP release of the Resume Tailoring Platform.

9.1 Included Features

Feature	Included	Notes
Resume Upload	Yes	Users upload a resume once.
Job Description Input	Yes	Paste JD text or use generic job page URL.
ATS Scoring	Yes	Fast ATS score generation.
Tailored Resume Generation	Yes	One-page optimized resume output.
Cover Letter Generation	Yes	Generated using resume and JD context.

PDF Export	Yes	Downloadable professional PDF output.
Auto Apply	No	Excluded from MVP scope.
Site-Specific Scraping	No	Only generic job page support is allowed.

10. Final Summary

The Resume Tailoring Platform MVP is designed as a lightweight, ATS-focused web application that helps users optimize resumes for specific job descriptions. The architecture prioritizes resume quality, one-page consistency, ATS relevance, AI-assisted improvements, and fast response time.

The system uses a modular service-based architecture with dedicated services for resume parsing, ATS scoring, AI generation, template formatting, and PDF export. This separation allows future scalability while keeping the MVP implementation simple and maintainable.

The platform follows a structured user flow: upload resume, provide job description, receive ATS insights, generate a tailored resume, generate a cover letter, and export the final result.

The selected technology stack balances simplicity, performance, and ease of implementation using Python, FastAPI, PostgreSQL, Next.js, AWS S3, and OpenAI API.